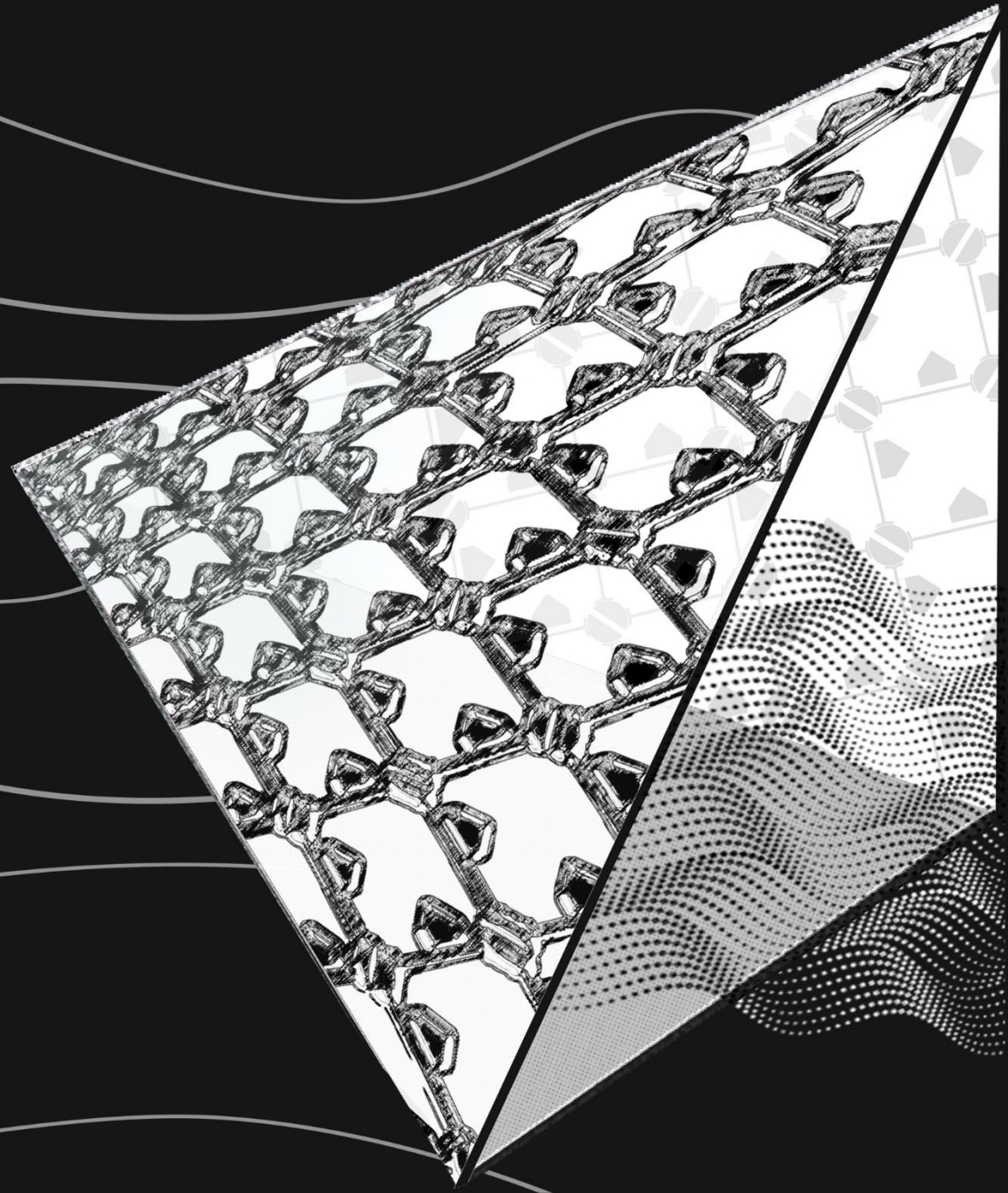


超导量子计算介绍

- Fei Yan (燕飞)
- 北京量子信息科学研究院(BAQIS)





NOBELPRISET I FYSIK 2025 THE NOBEL PRIZE IN PHYSICS 2025



KUNGL.
VETENSKAPS-
AKADEMIEN

THE ROYAL SWEDISH ACADEMY OF SCIENCES



Photo: University of California

John Clarke

University of California,
USA



Photo: Yale University

Michel H. Devoret

Yale University &
University of California, USA



Photo: Rocco Cavellin, Nature

John M. Martinis

University of California,
USA

"för upptäckten av makroskopisk kvantmekanisk tunnling och energikvantisering i en elektrisk krets"

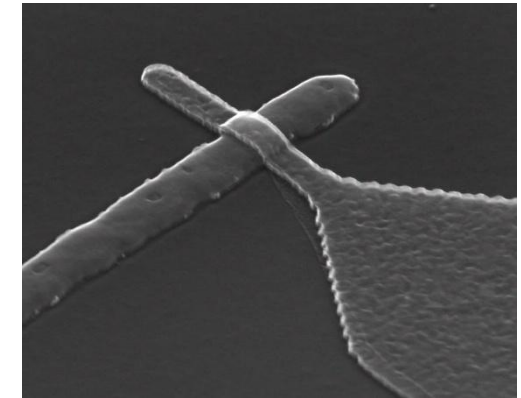
"for the discovery of macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit"

#NobelPrize

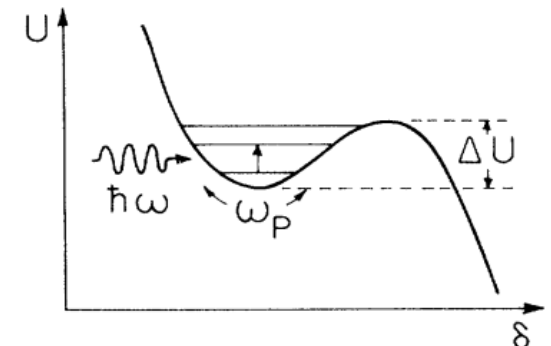
THE
NOBEL
PRIZE

Josephson junction

约瑟夫森结

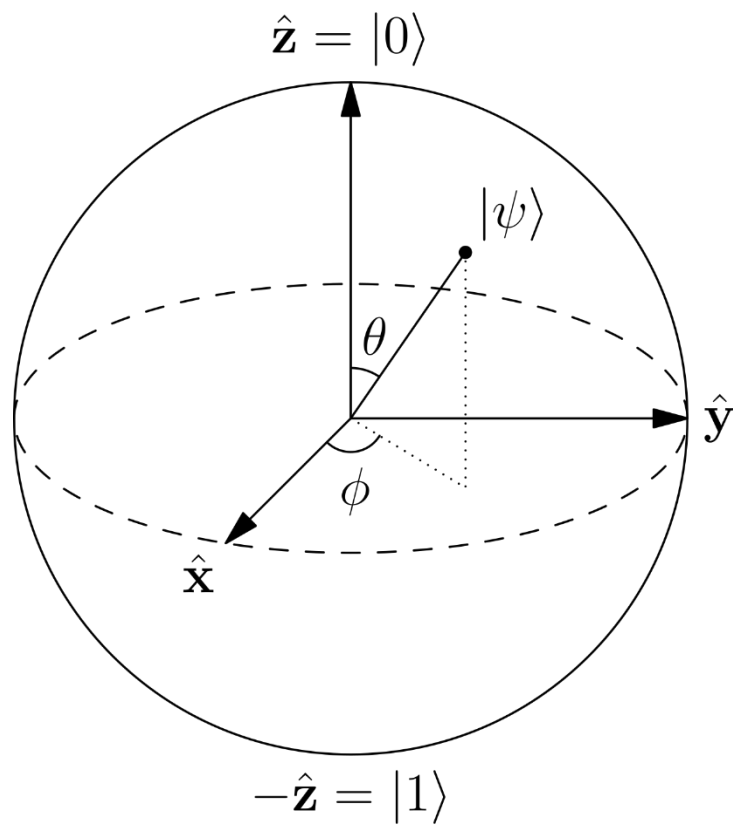


能级量子化
与量子隧穿



开启了人工量子系统、量子工程的新时代

Qubit primer



Superposition state

Define:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Superposition state:

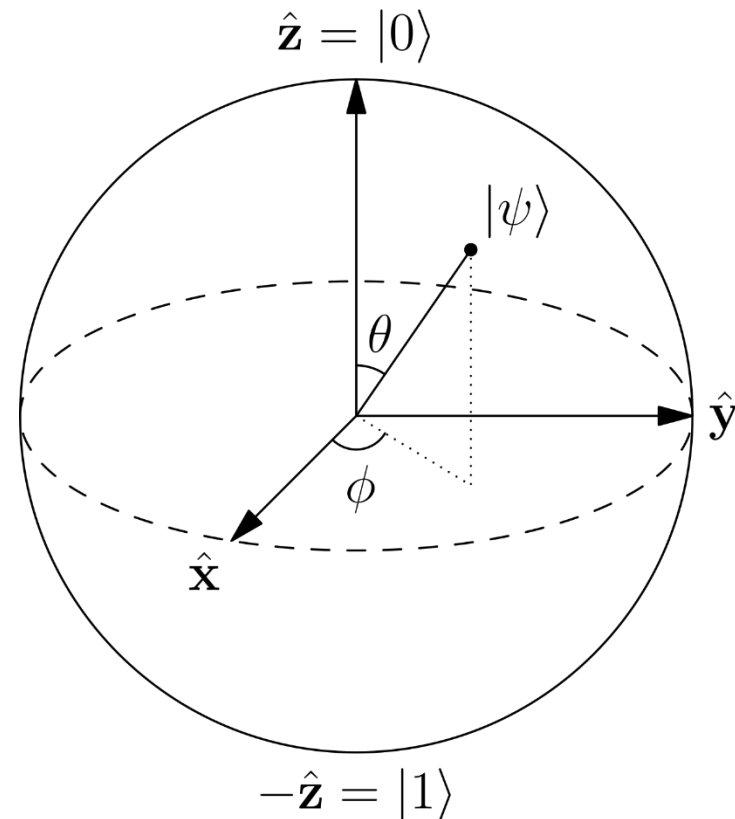
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$|\alpha|^2 + |\beta|^2 = 1$$

Bloch representation:

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \end{pmatrix}$$

Bloch sphere



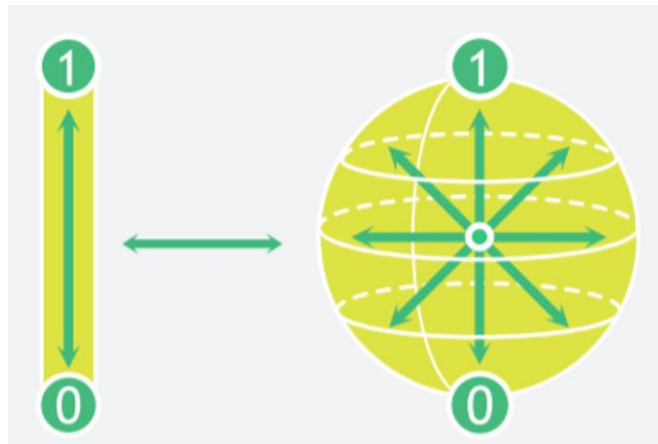
Superposition state

Classical bit

Quantum bit

either 0 or 1

0 and 1



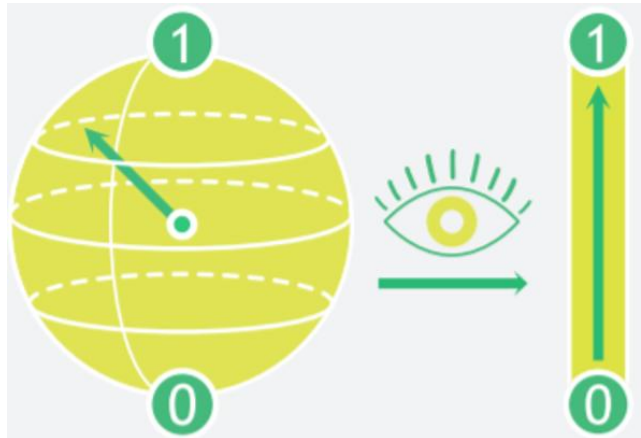
Measuring a qubit

Classical bit

either 0 or 1

Quantum bit

either 0 and 1,
probability depends on
pre-measurement state



More qubits

Classical bit

00 or 01 or 10 or 11

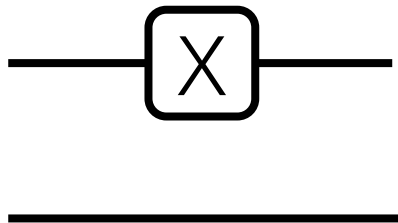
Quantum bit

$$|\psi\rangle = a_1|00\rangle + a_2|01\rangle + a_3|10\rangle + a_4|11\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}$$

What about 10 qubits? 100 qubits?

Quantum gate

NOT Gate



$|00\rangle \longrightarrow |10\rangle$

$|01\rangle \longrightarrow |11\rangle$

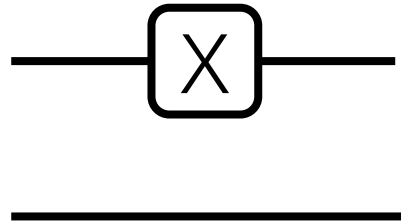
$|10\rangle \longrightarrow |00\rangle$

$|11\rangle \longrightarrow |01\rangle$

Quantum
Parallelism

Quantum gate

NOT Gate



$$|00\rangle \longrightarrow |10\rangle$$

$$|01\rangle \longrightarrow |11\rangle$$

$$|10\rangle \longrightarrow |00\rangle$$

$$|11\rangle \longrightarrow |01\rangle$$

Quantum
Parallelism

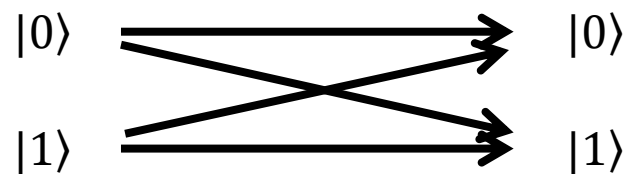
$$U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} \xrightarrow{U} \begin{pmatrix} a_3 \\ a_4 \\ a_1 \\ a_2 \end{pmatrix}$$

Quantum gate

Hadamard Gate



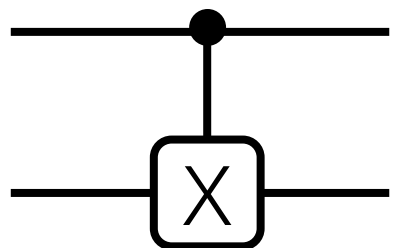
Quantum
Interference

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \xrightarrow{U} \begin{pmatrix} a_1 + a_2 \\ a_1 - a_2 \end{pmatrix}$$

Quantum gate

CNOT Gate



$$|00\rangle \longrightarrow |00\rangle$$

$$|01\rangle \longrightarrow |01\rangle$$

$$|10\rangle \longrightarrow |11\rangle$$

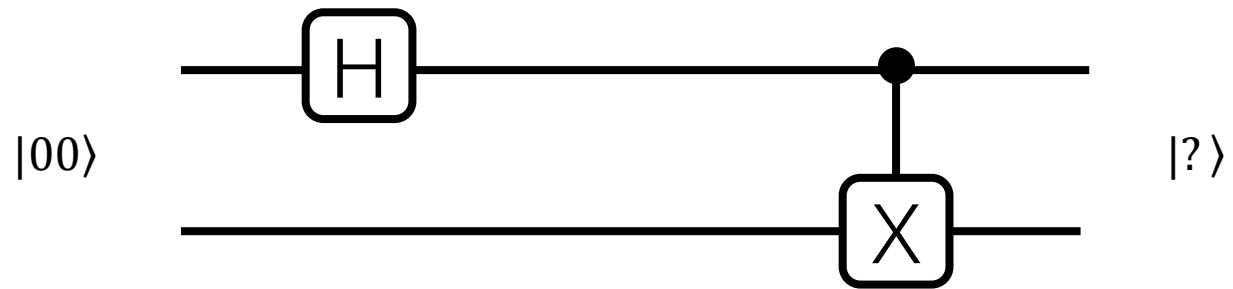
$$|11\rangle \longrightarrow |10\rangle$$

$$U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} \xrightarrow{U} \begin{pmatrix} a_1 \\ a_2 \\ a_4 \\ a_3 \end{pmatrix}$$

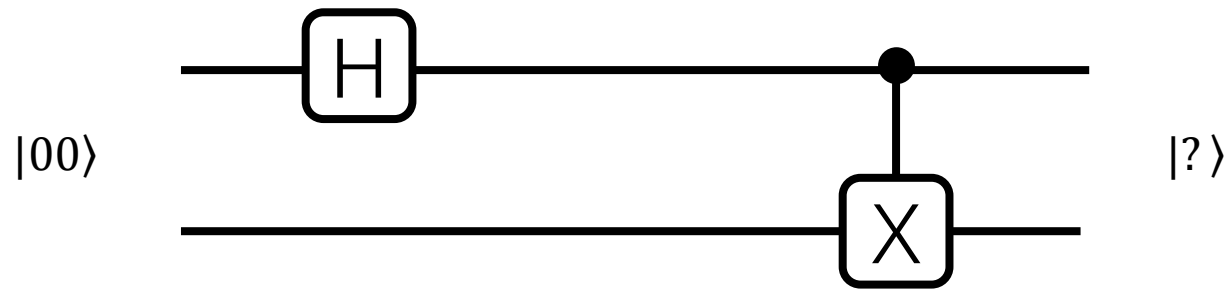
Quantum gate

CNOT Gate



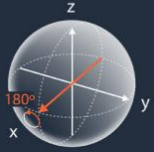
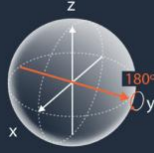
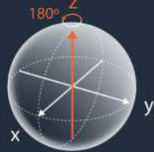
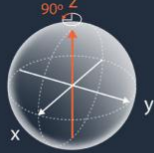
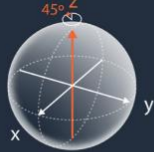
Quantum gate

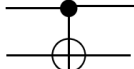
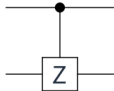
CNOT Gate



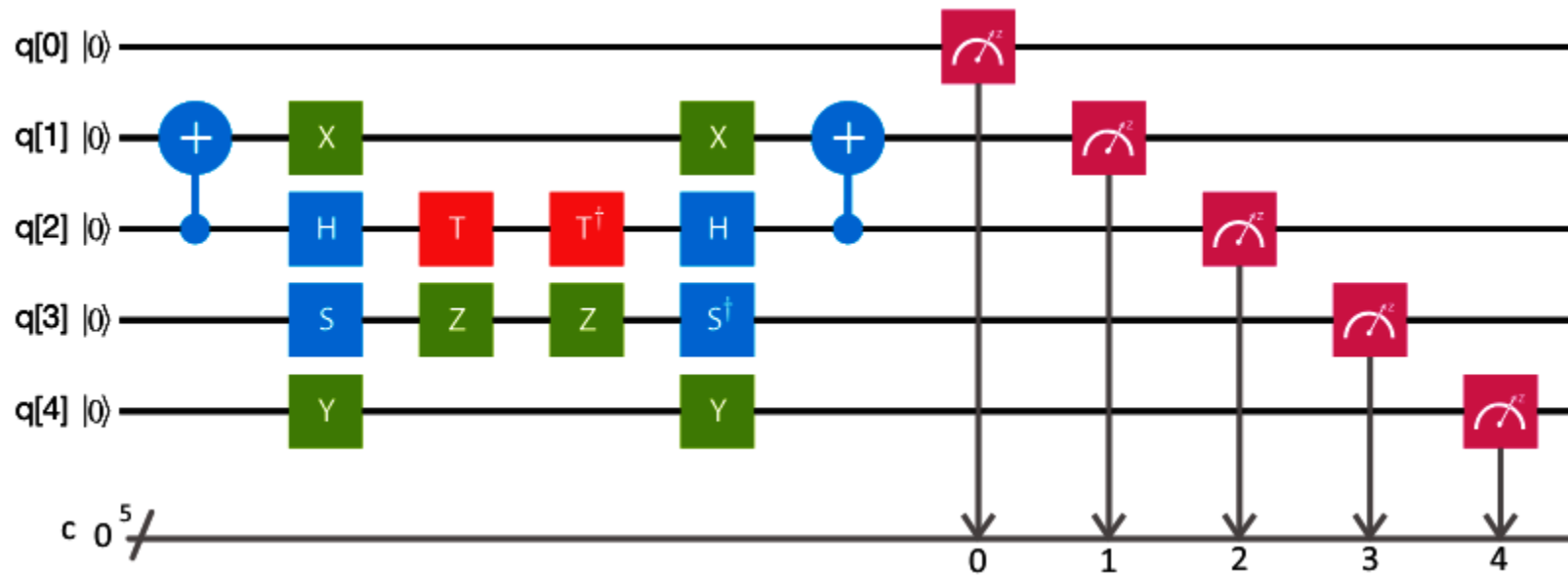
$$|00\rangle \longrightarrow \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) \longrightarrow \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Quantum
Entanglement

GATE	CIRCUIT REPRESENTATION	MATRIX REPRESENTATION	TRUTH TABLE	BLOCH SPHERE						
I Identity-gate: no rotation is performed.		$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td> 0⟩</td><td> 0⟩</td></tr><tr><td> 1⟩</td><td> 1⟩</td></tr></table>	Input	Output	0⟩	0⟩	1⟩	1⟩	
Input	Output									
0⟩	0⟩									
1⟩	1⟩									
X gate: rotates the qubit state by π radians (180°) about the x-axis.		$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td> 0⟩</td><td> 1⟩</td></tr><tr><td> 1⟩</td><td> 0⟩</td></tr></table>	Input	Output	0⟩	1⟩	1⟩	0⟩	
Input	Output									
0⟩	1⟩									
1⟩	0⟩									
Y gate: rotates the qubit state by π radians (180°) about the y-axis.		$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td> 0⟩</td><td>i 1⟩</td></tr><tr><td> 1⟩</td><td>-i 0⟩</td></tr></table>	Input	Output	0⟩	i 1⟩	1⟩	-i 0⟩	
Input	Output									
0⟩	i 1⟩									
1⟩	-i 0⟩									
Z gate: rotates the qubit state by π radians (180°) about the z-axis.		$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td> 0⟩</td><td> 0⟩</td></tr><tr><td> 1⟩</td><td>- 1⟩</td></tr></table>	Input	Output	0⟩	0⟩	1⟩	- 1⟩	
Input	Output									
0⟩	0⟩									
1⟩	- 1⟩									
S gate: rotates the qubit state by $\frac{\pi}{2}$ radians (90°) about the z-axis.		$S = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td> 0⟩</td><td> 0⟩</td></tr><tr><td> 1⟩</td><td>$e^{i\frac{\pi}{2}} 1\rangle$</td></tr></table>	Input	Output	0⟩	0⟩	1⟩	$e^{i\frac{\pi}{2}} 1\rangle$	
Input	Output									
0⟩	0⟩									
1⟩	$e^{i\frac{\pi}{2}} 1\rangle$									
T gate: rotates the qubit state by $\frac{\pi}{4}$ radians (45°) about the z-axis.		$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td> 0⟩</td><td> 0⟩</td></tr><tr><td> 1⟩</td><td>$e^{i\frac{\pi}{4}} 1\rangle$</td></tr></table>	Input	Output	0⟩	0⟩	1⟩	$e^{i\frac{\pi}{4}} 1\rangle$	
Input	Output									
0⟩	0⟩									
1⟩	$e^{i\frac{\pi}{4}} 1\rangle$									
H gate: rotates the qubit state by π radians (180°) about an axis diagonal in the x-z plane. This is equivalent to an X-gate followed by a $\frac{\pi}{2}$ rotation about the y-axis.		$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td> 0⟩</td><td>$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$</td></tr><tr><td> 1⟩</td><td>$\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$</td></tr></table>	Input	Output	0⟩	$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$	1⟩	$\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$	
Input	Output									
0⟩	$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$									
1⟩	$\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$									

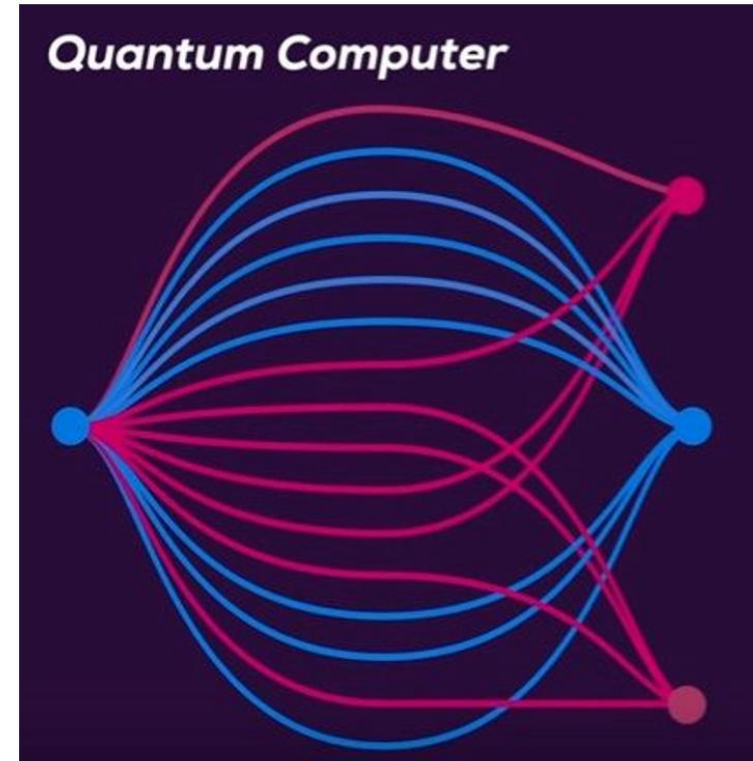
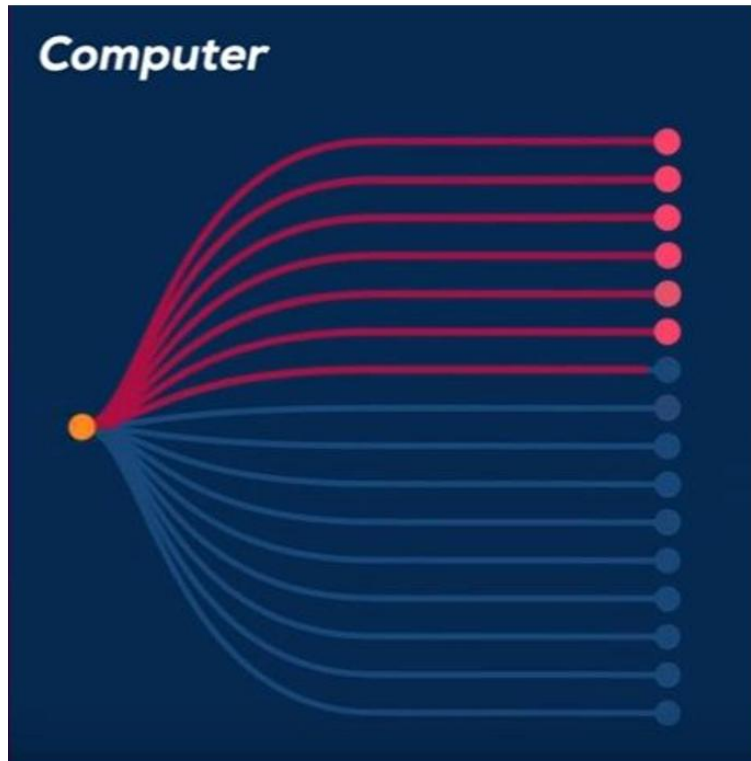
GATE	CIRCUIT REPRESENTATION	MATRIX REPRESENTATION	TRUTH TABLE										
Controlled-NOT gate: apply an X-gate to the target qubit if the control qubit is in state $ 1\rangle$		$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$	<table><thead><tr><th>Input</th><th>Output</th></tr></thead><tbody><tr><td>$00\rangle$</td><td>$00\rangle$</td></tr><tr><td>$01\rangle$</td><td>$01\rangle$</td></tr><tr><td>$10\rangle$</td><td>$11\rangle$</td></tr><tr><td>$11\rangle$</td><td>$10\rangle$</td></tr></tbody></table>	Input	Output	$ 00\rangle$	$ 00\rangle$	$ 01\rangle$	$ 01\rangle$	$ 10\rangle$	$ 11\rangle$	$ 11\rangle$	$ 10\rangle$
Input	Output												
$ 00\rangle$	$ 00\rangle$												
$ 01\rangle$	$ 01\rangle$												
$ 10\rangle$	$ 11\rangle$												
$ 11\rangle$	$ 10\rangle$												
Controlled-phase gate: apply a Z-gate to the target qubit if the control qubit is in state $ 1\rangle$		$\text{CPHASE} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$	<table><thead><tr><th>Input</th><th>Output</th></tr></thead><tbody><tr><td>$00\rangle$</td><td>$00\rangle$</td></tr><tr><td>$01\rangle$</td><td>$01\rangle$</td></tr><tr><td>$10\rangle$</td><td>$10\rangle$</td></tr><tr><td>$11\rangle$</td><td>$- 11\rangle$</td></tr></tbody></table>	Input	Output	$ 00\rangle$	$ 00\rangle$	$ 01\rangle$	$ 01\rangle$	$ 10\rangle$	$ 10\rangle$	$ 11\rangle$	$- 11\rangle$
Input	Output												
$ 00\rangle$	$ 00\rangle$												
$ 01\rangle$	$ 01\rangle$												
$ 10\rangle$	$ 10\rangle$												
$ 11\rangle$	$- 11\rangle$												

Quantum circuit / algorithm



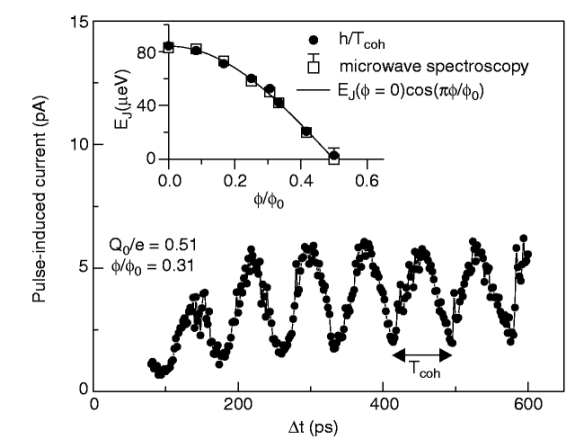
Quantum
Circuit

Quantum circuit / algorithm



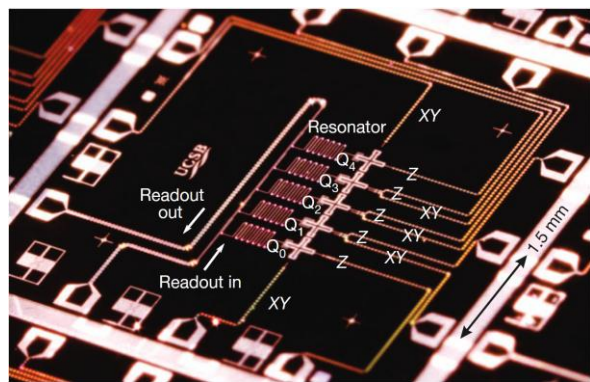
Superconducting qubit : History

首次观测到量子相干震荡(NEC)



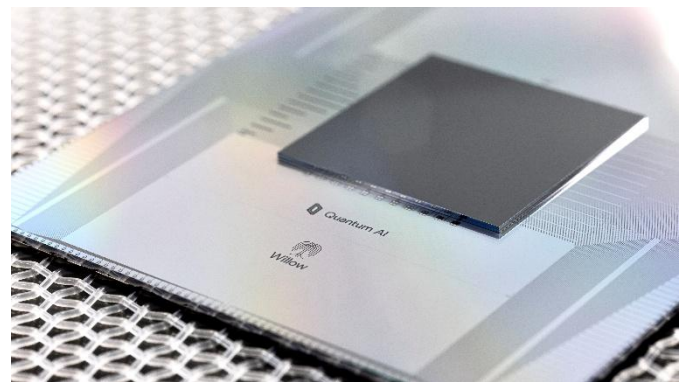
Nature 398,786–788 (1999)

错误率低于纠错阈值(UCSB)



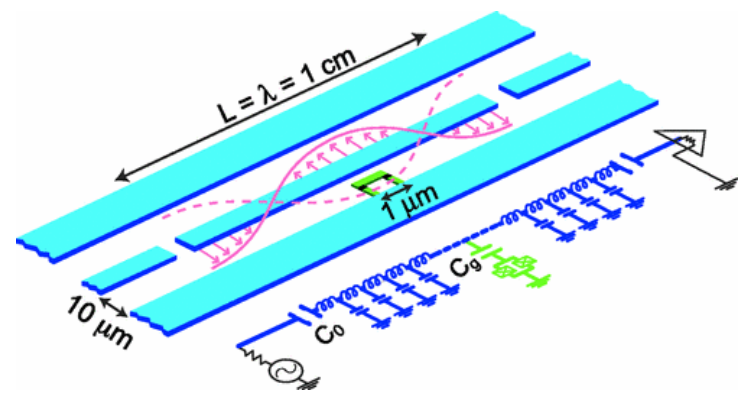
Nature 508,500-503 (2014)

量子纠错获得增益(Google)



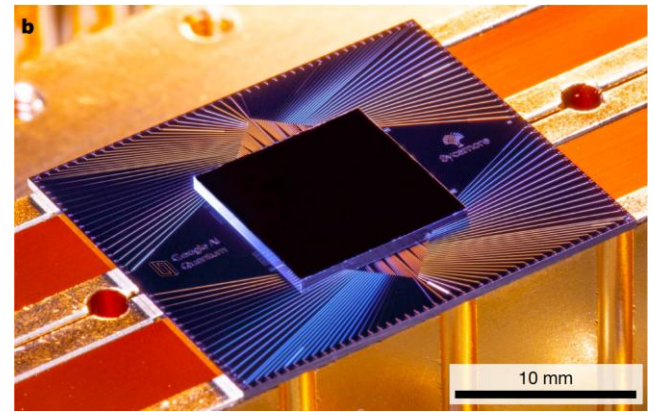
Nature 638,920–926 (2025)

CircuitQED架构诞生(Yale)



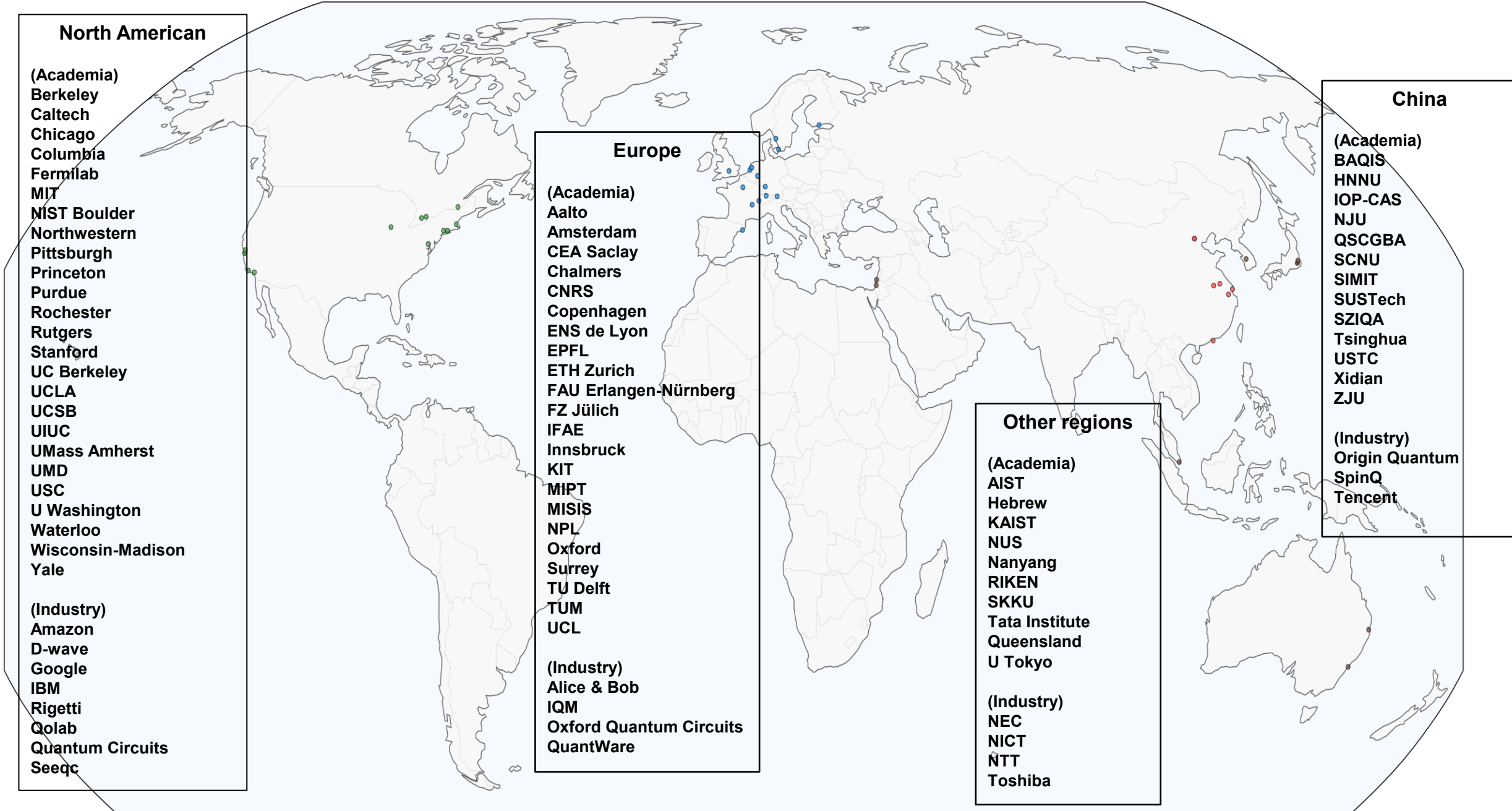
Phys. Rev.A 69, 062320 (2004)

量子优越性展示(Google)



Nature 574,505–510 (2019)

Superconducting qubit : World map

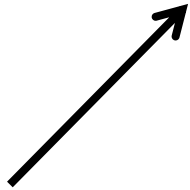


Superconducting qubit:

Superconducting Qubit

= Superconducting Quantum Circuit

Superconducting Qubit = Superconducting Quantum Circuit



Material platform:

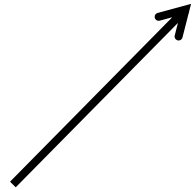
Superconducting

- Al, Nb, Ta
- Dissipationless
- Josephson junction

材料：超导

- 无耗散
- 约瑟夫森结

Superconducting Qubit = Superconducting Quantum Circuit



Material platform:

Superconducting

- Al, Nb, Ta
- Dissipationless
- Josephson junction

材料：超导

- 无耗散
- 约瑟夫森结



Working condition:

Quantum regime

- Low temperature ($\sim 10\text{mK}$)
- Low-noise environment

工作条件：量子极限

- 低温、低噪声

Superconducting Qubit = Superconducting Quantum Circuit

Material platform:

Superconducting

- Al, Nb, Ta
- Dissipationless
- Josephson junction

材料：超导

- 无耗散
- 约瑟夫森结

Working condition:

Quantum regime

- Low temperature ($\sim 10\text{mK}$)
- Low-noise environment

工作条件：量子极限

- 低温、低噪声

Structural form:

Microwave circuit

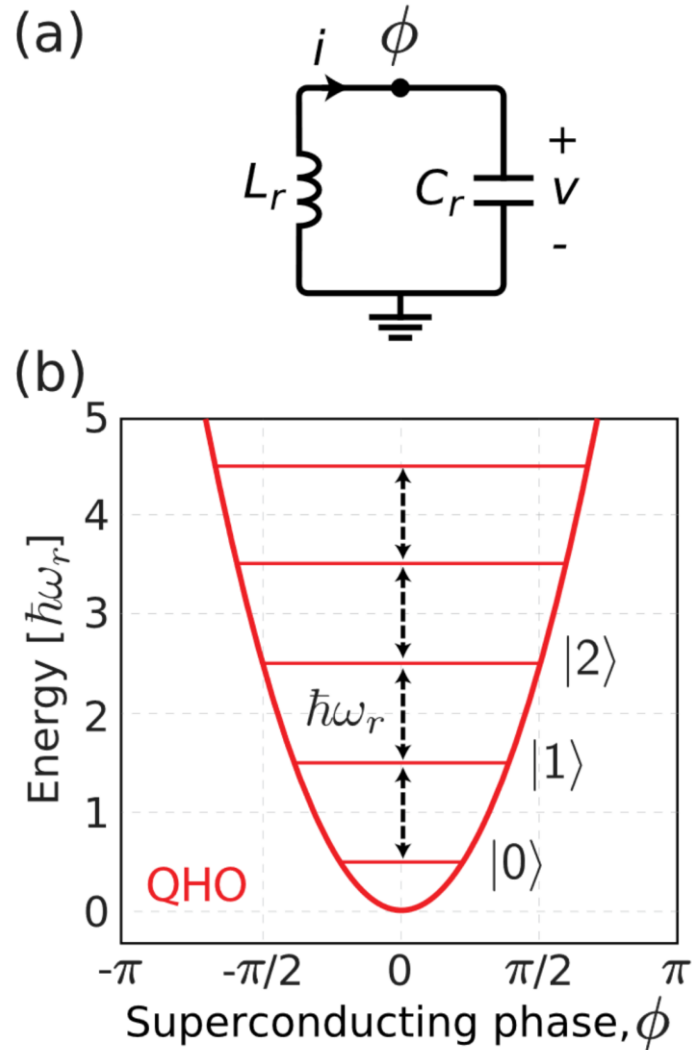
- Nonlinear LC resonator
- 4-8GHz

形式：电路

- 微波电路结构
- 非线性谐振器

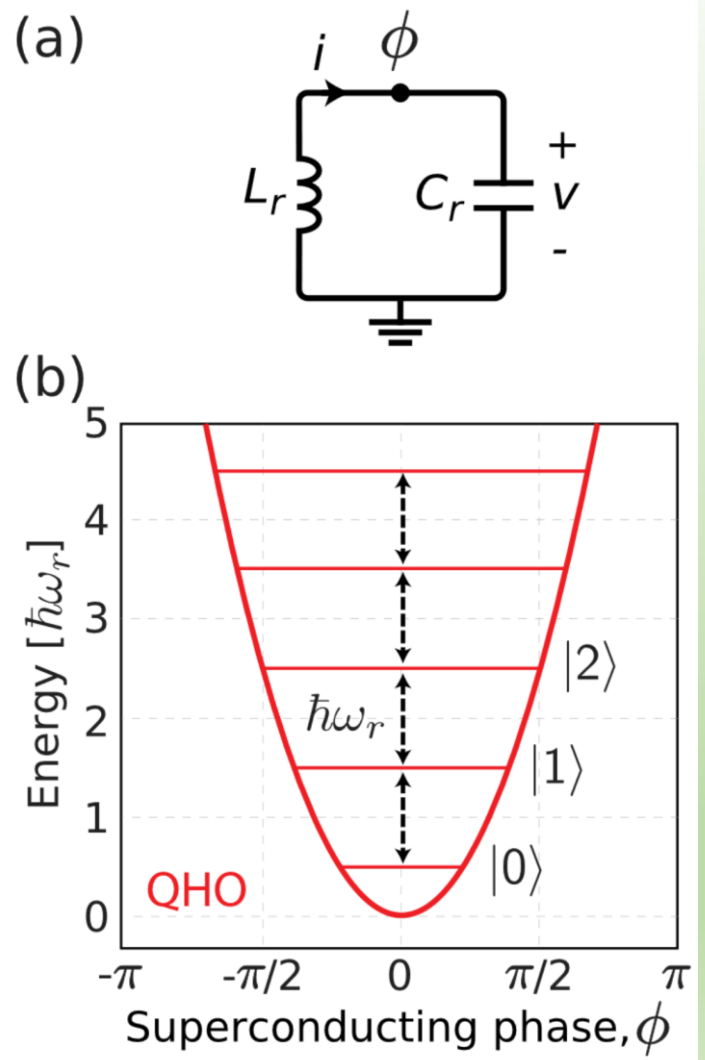
Superconducting qubit:

简谐振子

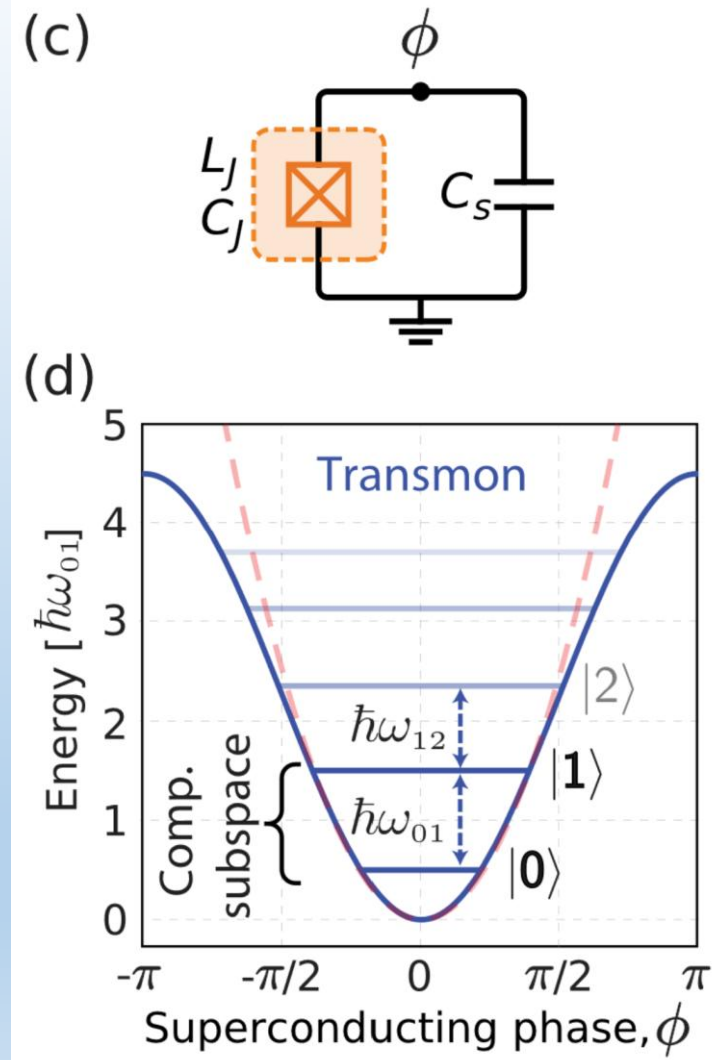


Superconducting qubit:

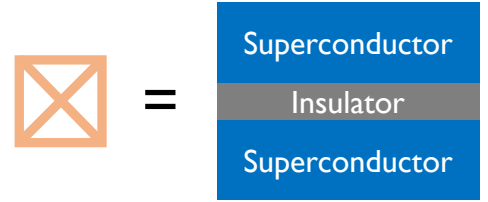
简谐振子



非谐振子



Josephson junction
~ nonlinear inductor

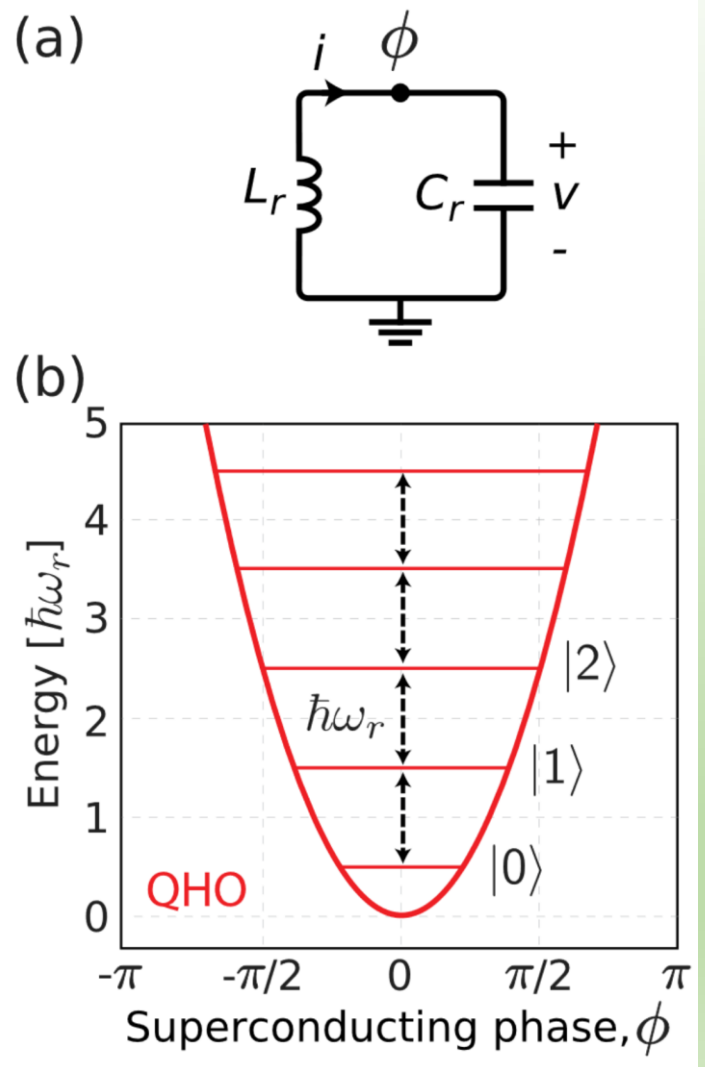


约瑟夫森结:

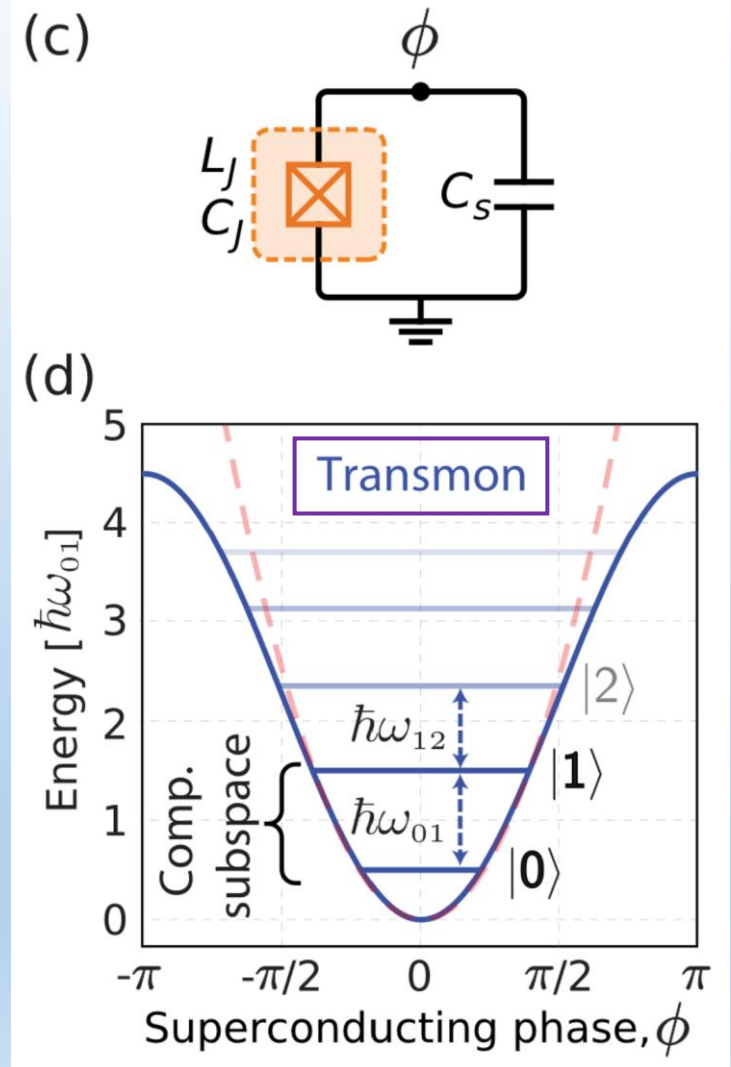
- 相当于非线性电感
- 并联电容形成非谐振子(Transmon量子比特)

Superconducting qubit:

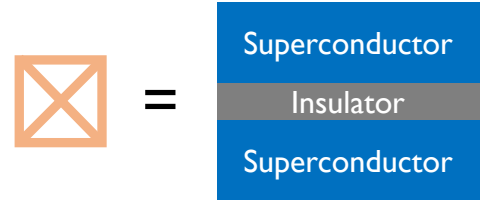
简谐振子



非谐振子



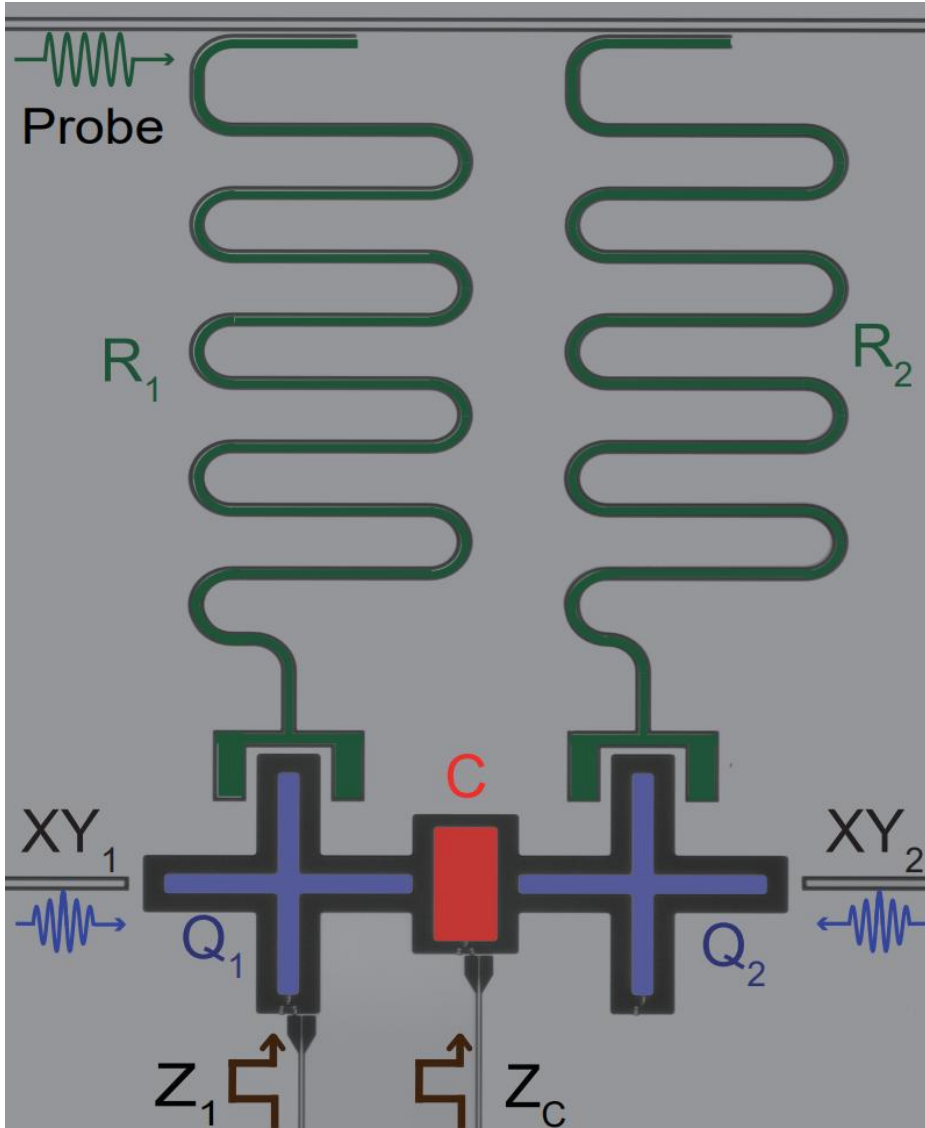
Josephson junction
~ nonlinear inductor



约瑟夫森结:

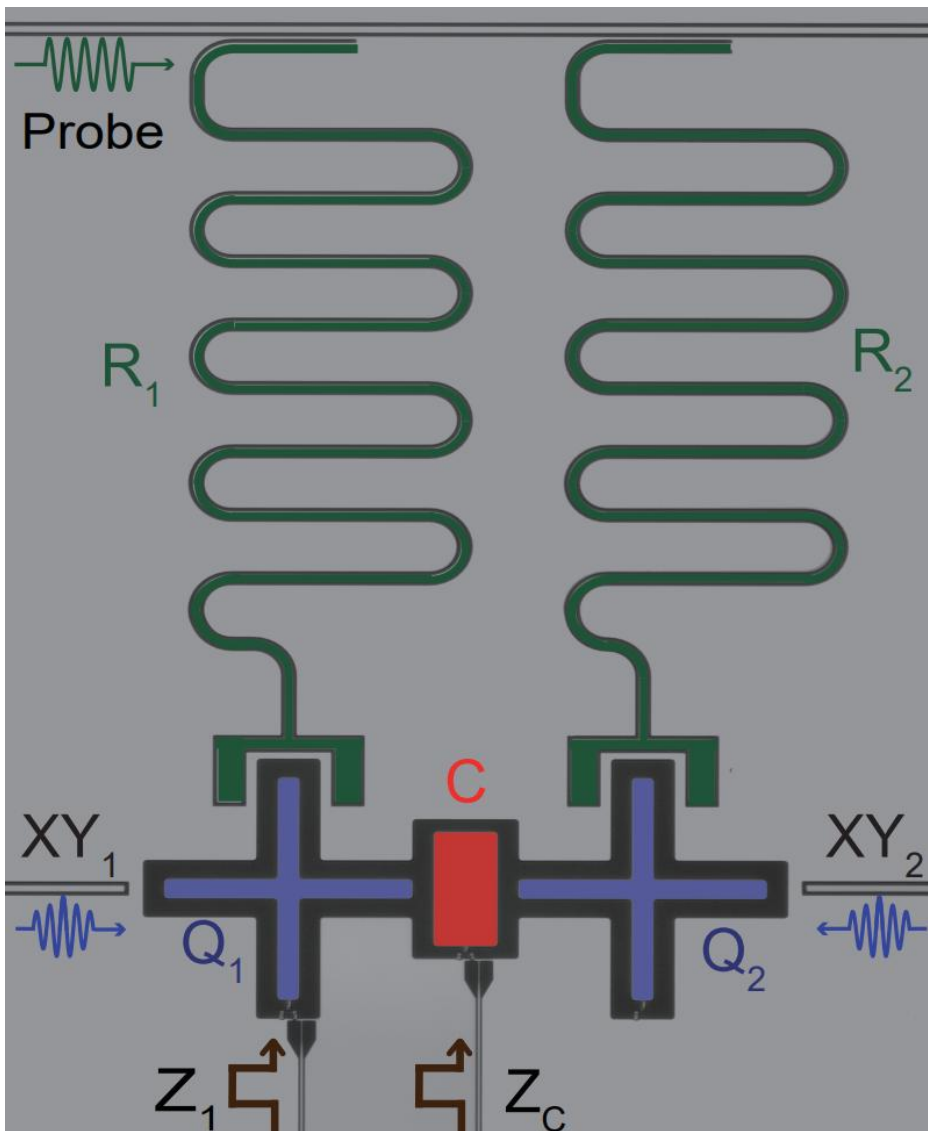
- 相当于非线性电感
- 并联电容形成非谐振子(Transmon量子比特)

Superconducting qubit:



- Readout resonators 读取谐振腔
- Qubits (Transmon) 量子比特
- Tunable coupler (Transmon) 可调耦合器
- Readout feedline 读取线
- Charge drive lines (XY) 驱动线
- Flux drive lines (Z) 磁通线

Superconducting qubit:



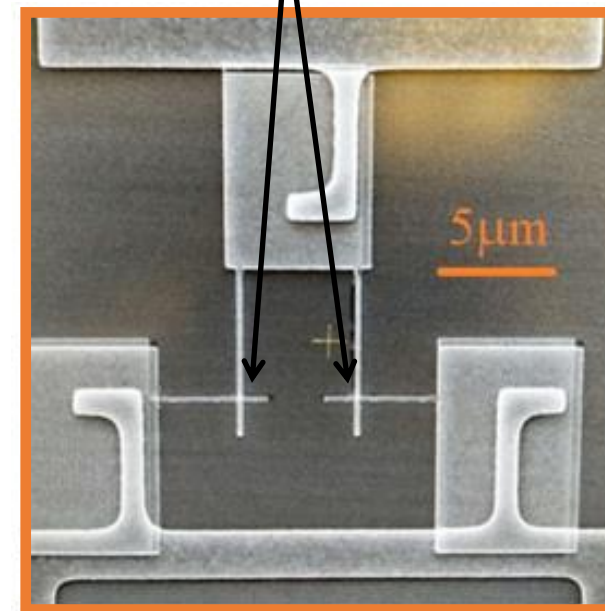
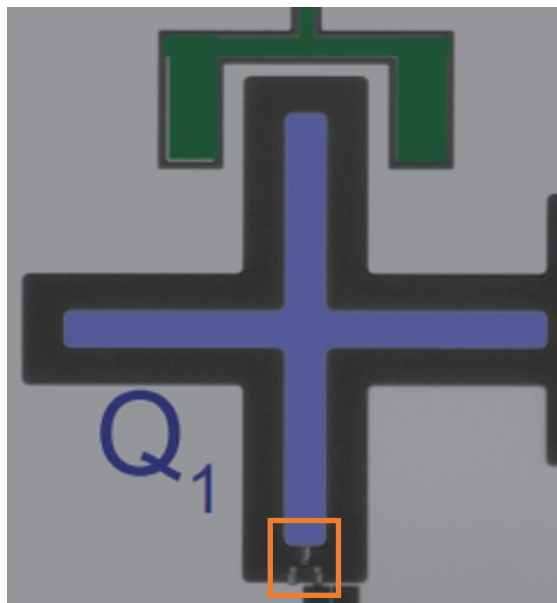
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- Charge drive lines (XY) 驱动线
- Flux drive lines (Z) 磁通线



=

Superconductor
Insulator
Superconductor

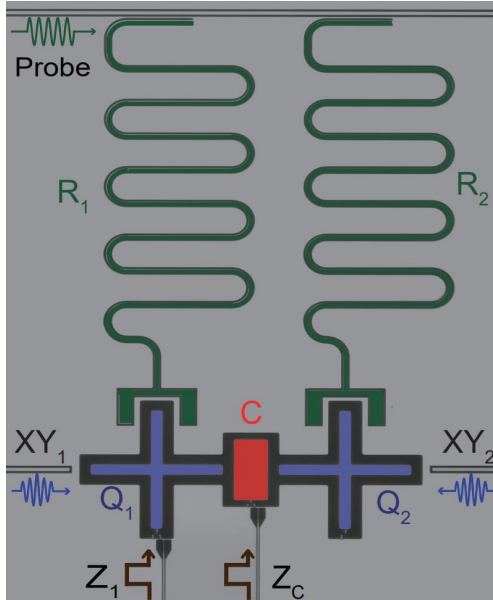
约瑟夫森结



Superconducting quantum computer

超导量子芯片

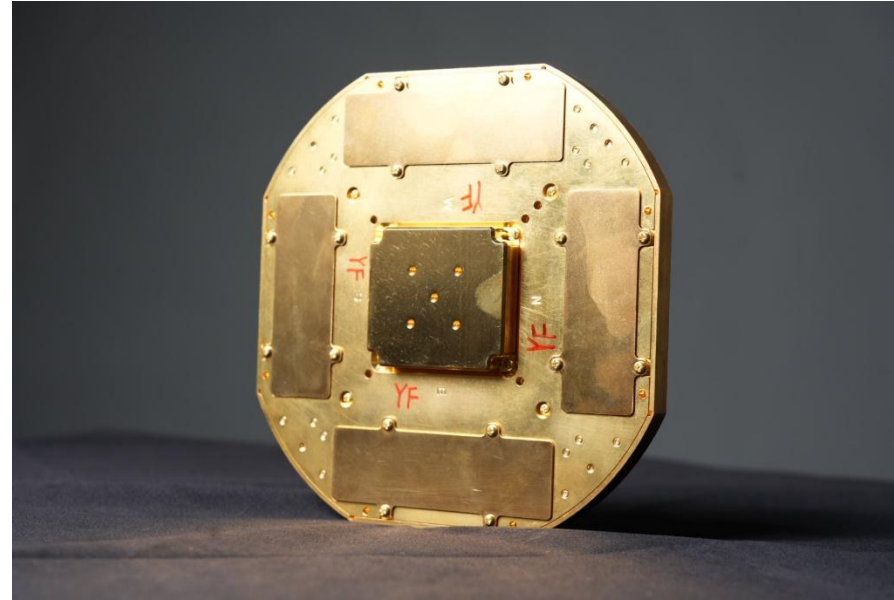
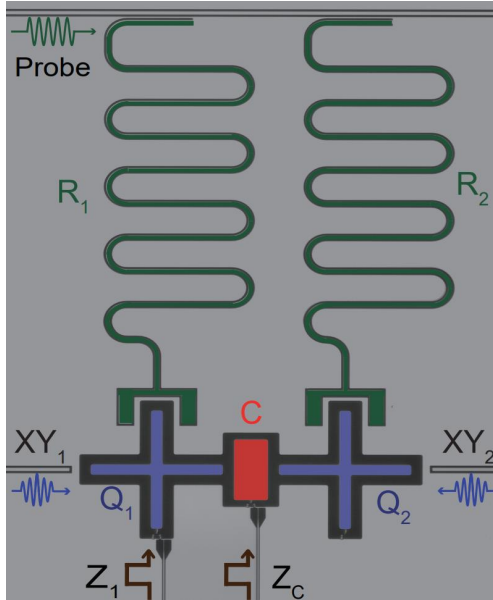
- Coherence
- Crosstalk



Superconducting quantum computer

超导量子芯片

- Coherence
- Crosstalk



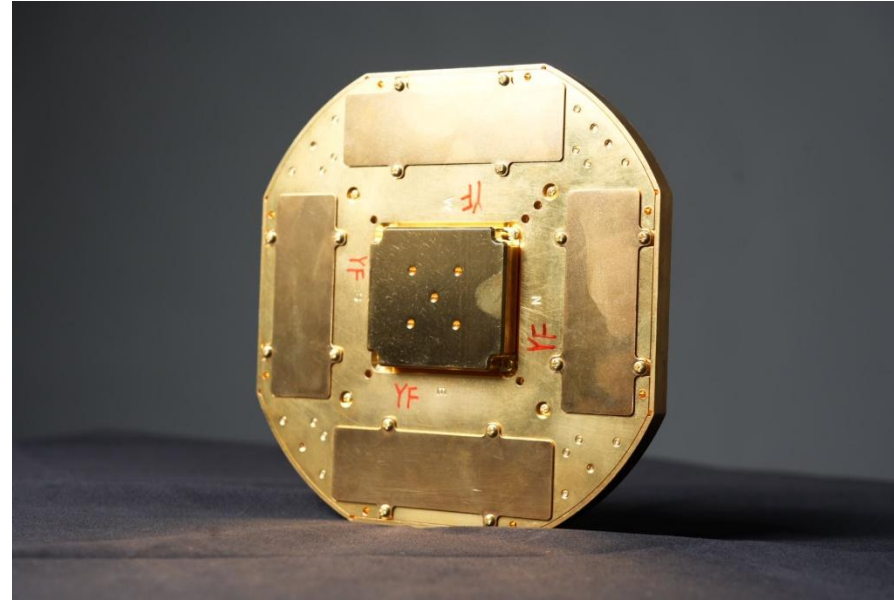
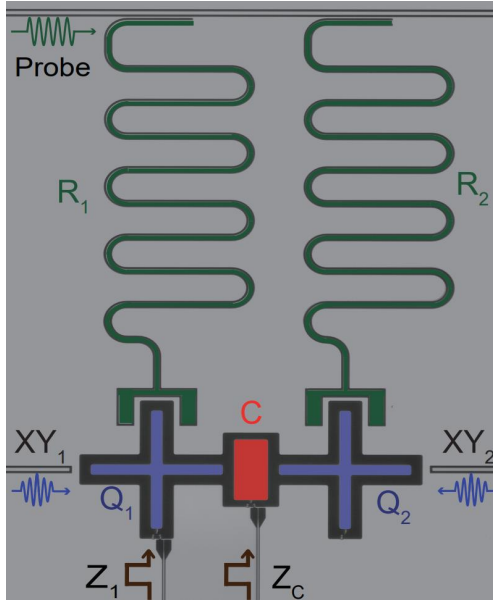
芯片封装盒

- EM Shielding
- Thermal anchoring

Superconducting quantum computer

超导量子芯片

- Coherence
- Crosstalk



芯片封装盒

- EM Shielding
- Thermal anchoring



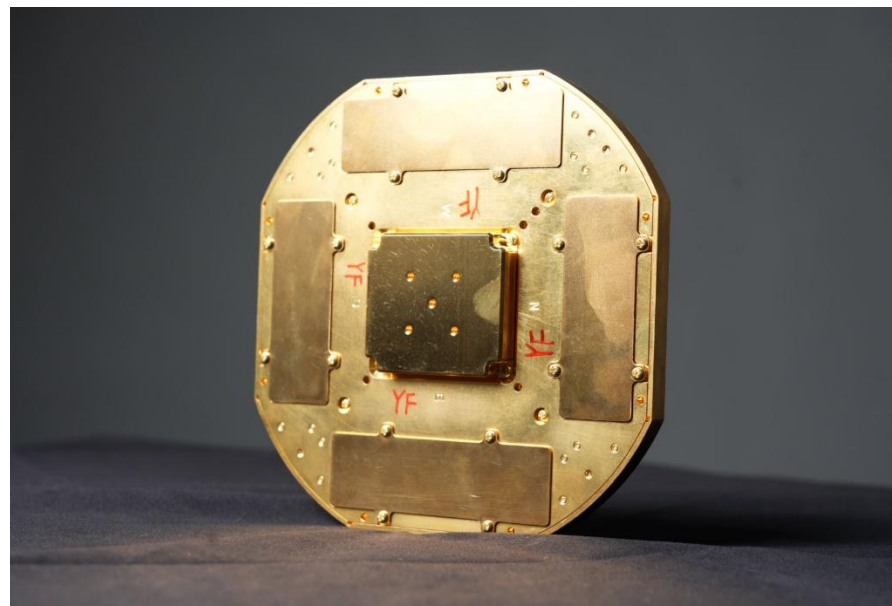
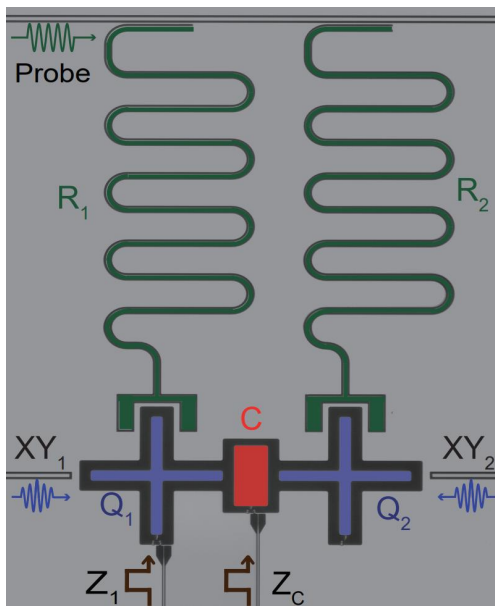
稀释制冷机

- Low temperature
- Noise attenuation

Superconducting quantum computer

超导量子芯片

- Coherence
- Crosstalk



芯片封装盒

- EM Shielding
- Thermal anchoring



稀释制冷机

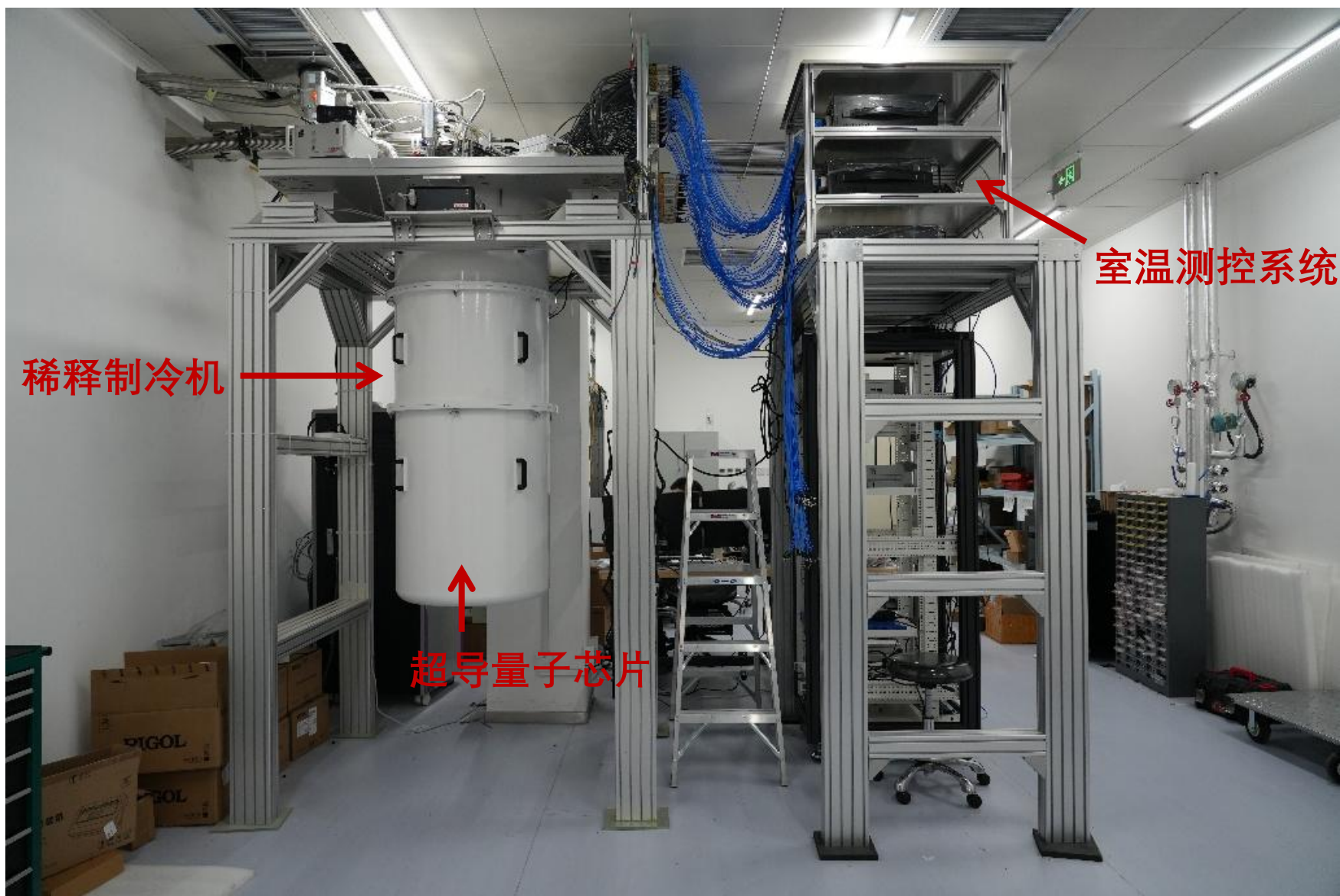
- Low temperature
- Noise attenuation



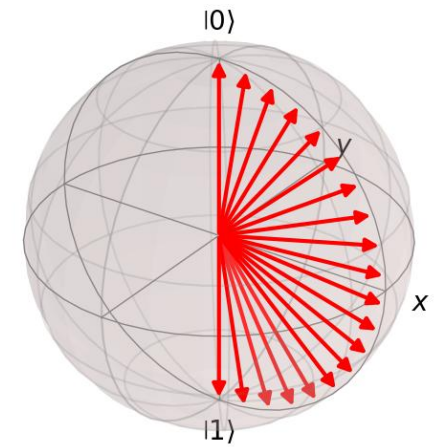
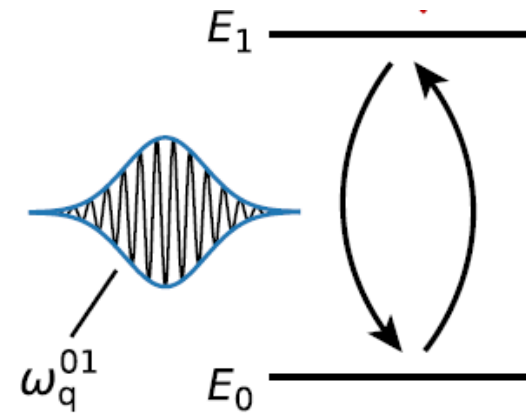
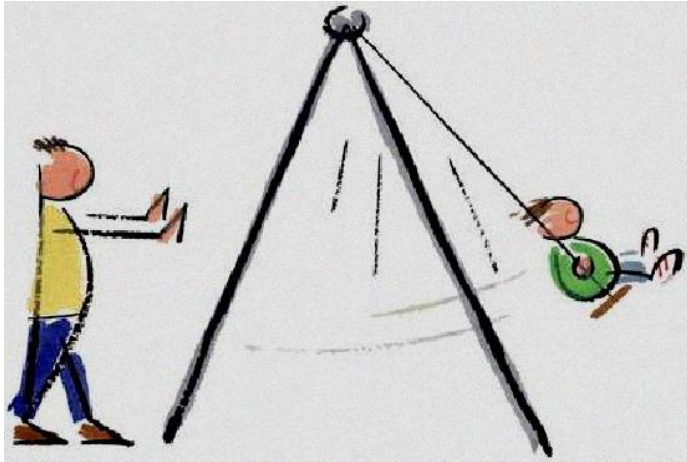
室温测控系统

- Precision control
- Low noise





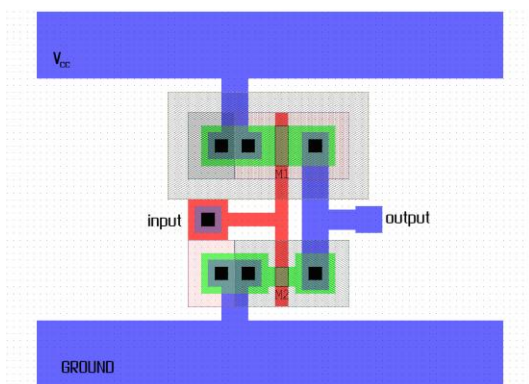
Superconducting qubit: Control



X gate

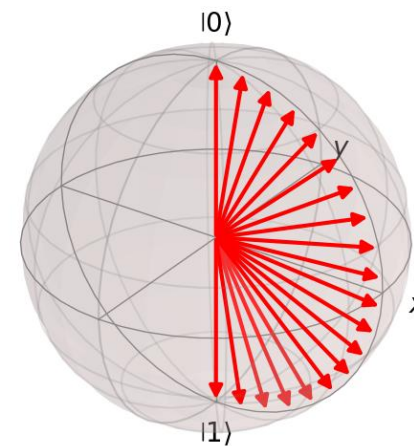
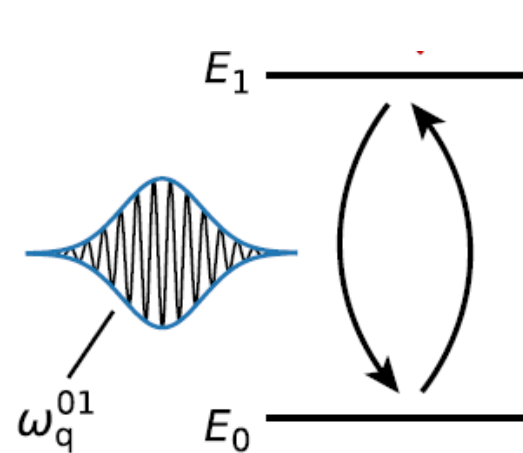
Superconducting qubit: Control

经典逻辑门(硬)



NOT gate

量子逻辑门(软)



X gate

量子逻辑门操作：复杂系统的控制

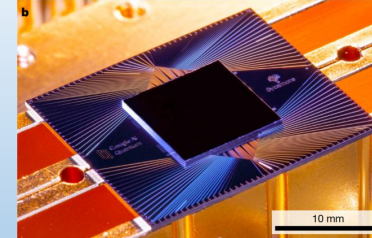
Quantum gate 量子门:

Classical bit



0.0000000000000001

Quantum bit



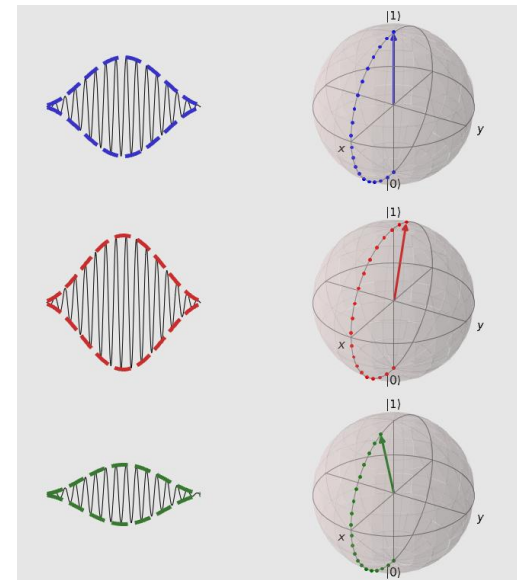
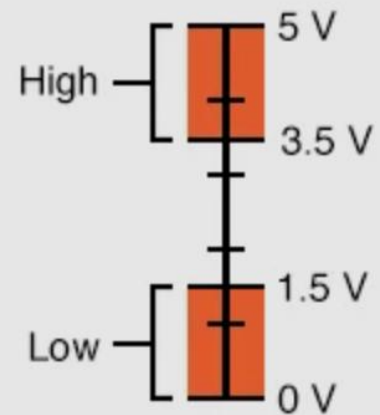
0.001-0.01

Gate error rate:
(Fidelity = $1 - \text{Error}$)

Digital operation

- Transistor (nonlinear)
- Large noise margin

Acceptable CMOS Gate
Input Signal Levels



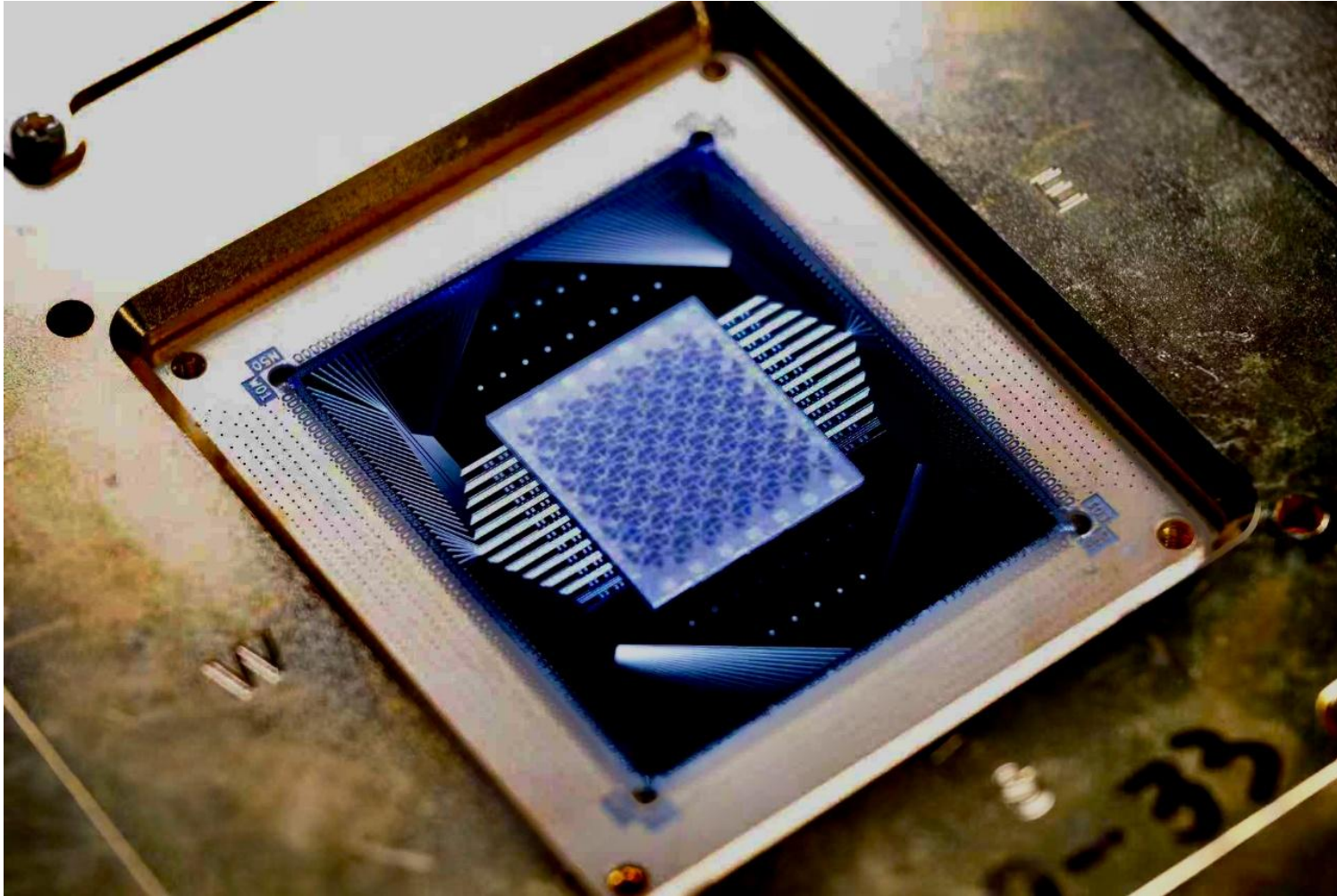
量子比特极易出错

- 环境噪声
- 控制不完美

Analog operation

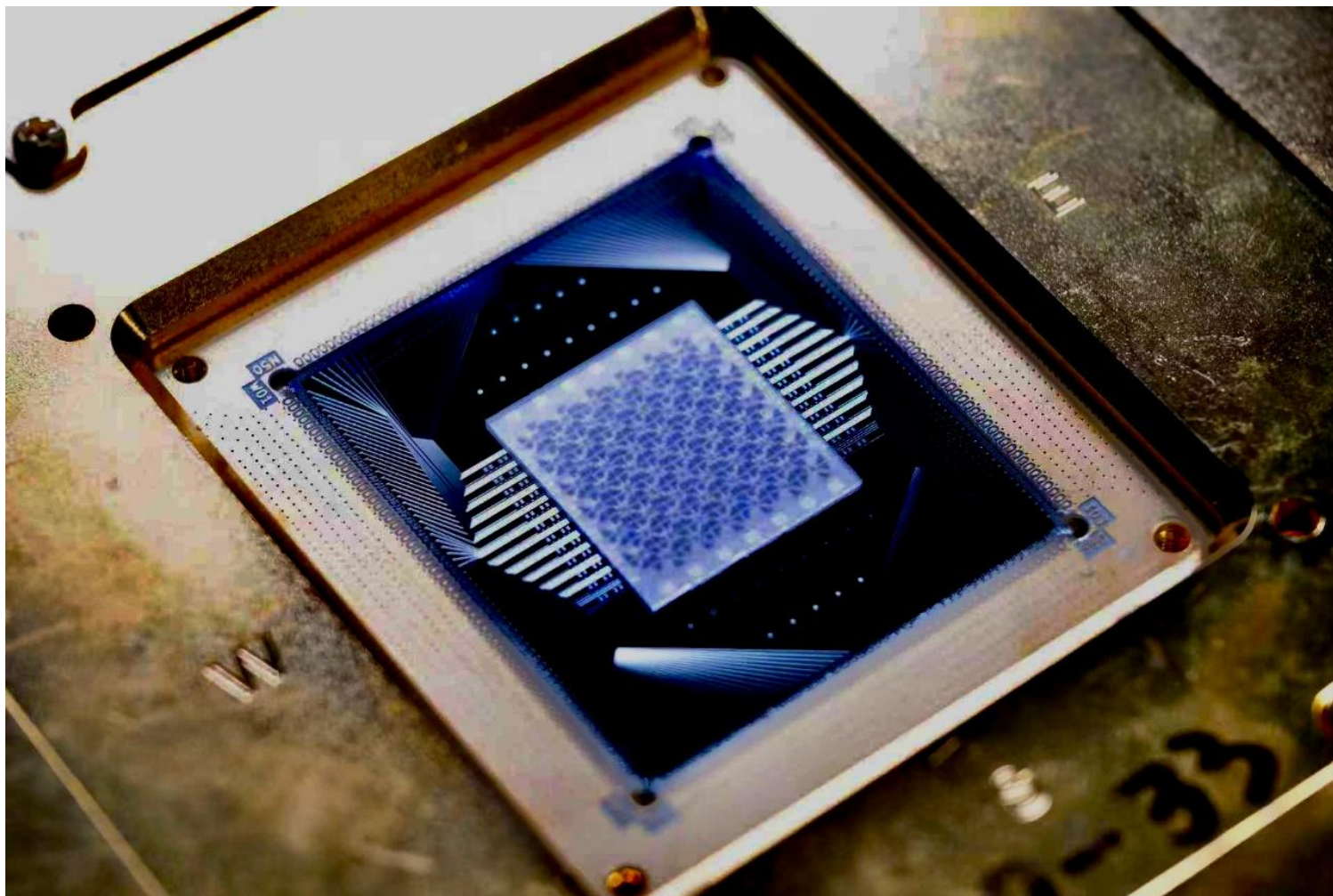
- Quantum dynamics (linear)
- No noise margin

Scaling efforts

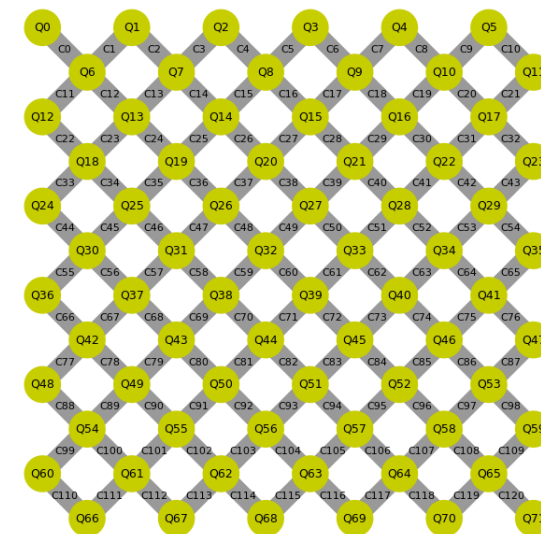


72-qubit processor

Scaling efforts



72-qubit processor

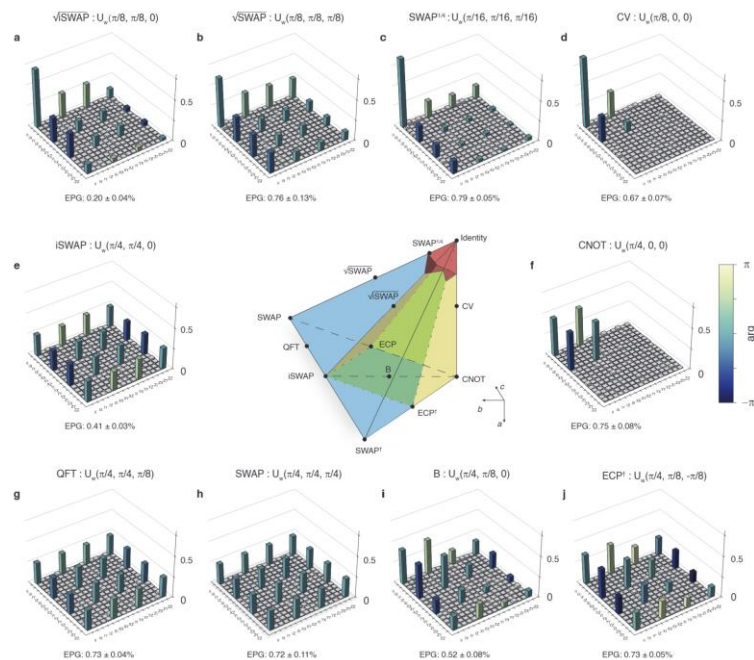


平面网格结构
同表面码兼容

Chip info:

- 72 transmon qubits
- 121 tunable couplers
- Square lattice
- (top) tantalum on sapphire
- (bottom) tantalum on silicon
- All-floating design

Arbitrary 2-qubit gate

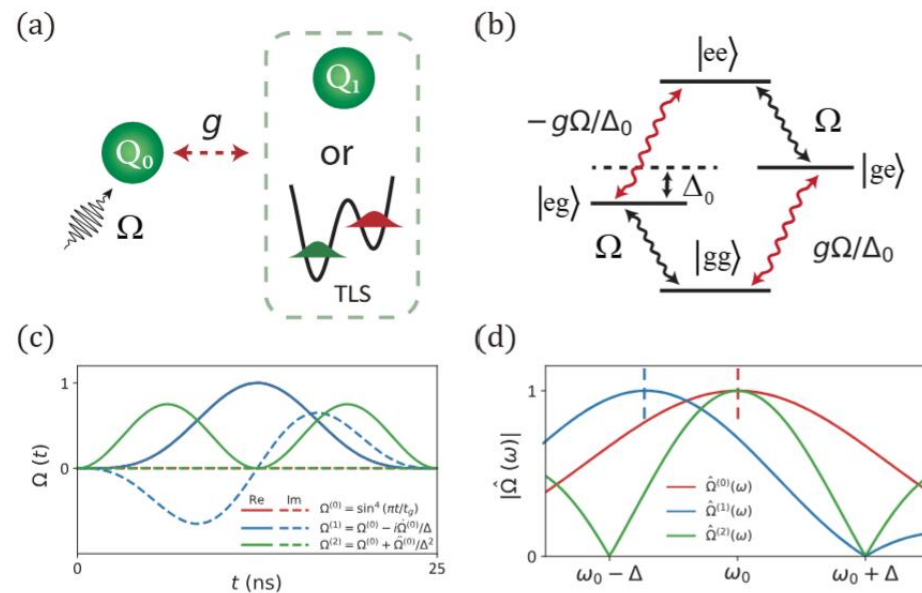


Nature Physics 21, 1489–1496 (2025)

硬件原生的任意两比特门

In collaboration with
Prof. Jianxin Chen (THU)

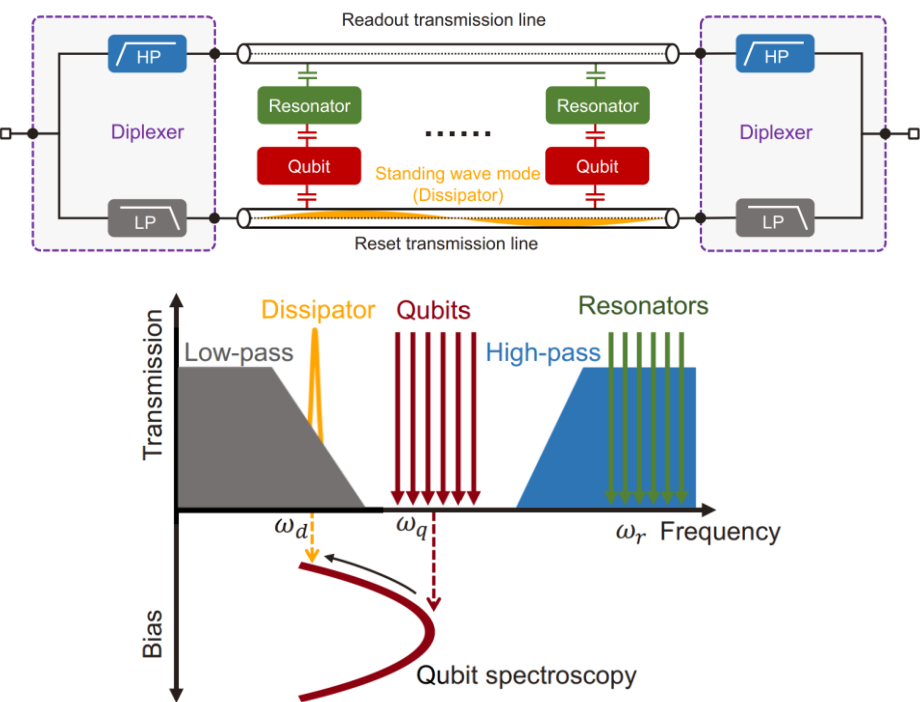
Suppressing spurious transition



Phys. Rev. Lett. 135, 160804 (2025)

抑制门操作中的杂散跃迁

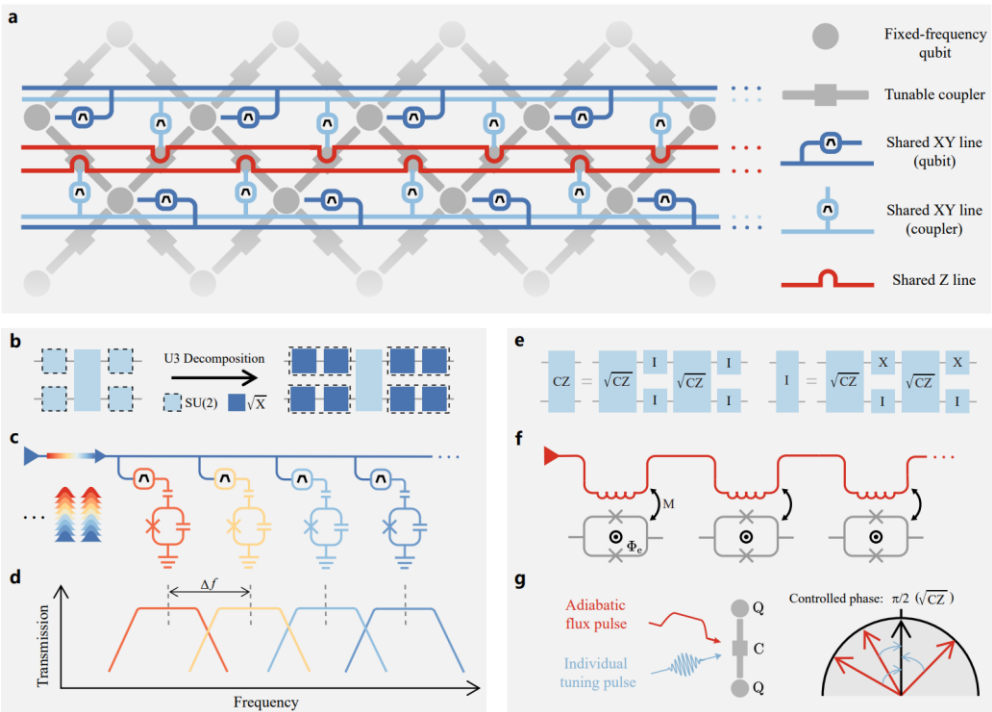
Fast reset and protective readout



Phys. Rev. Applied 23, 014012 (2025)

支持快速态重置和受保护读取新架构

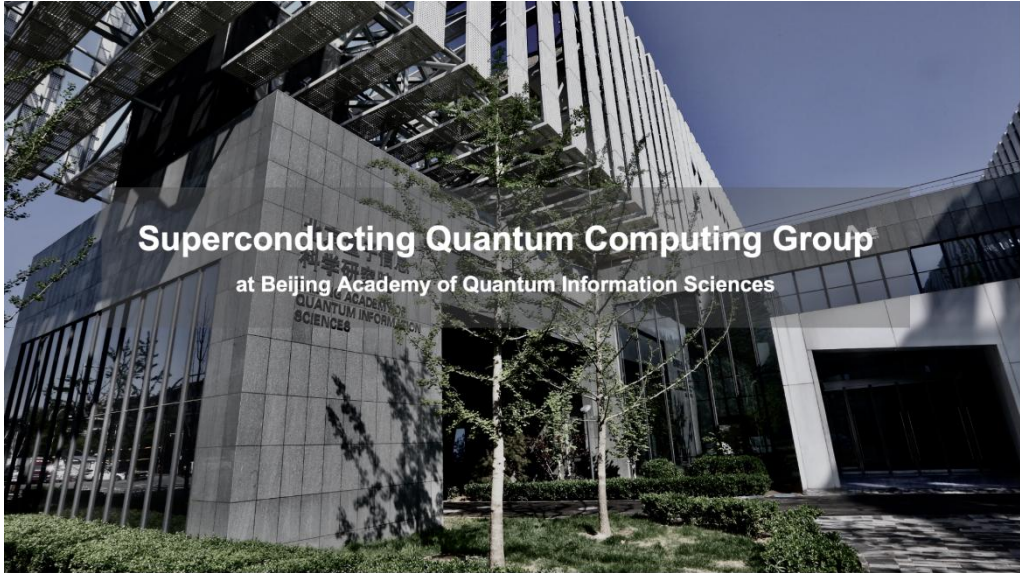
Multiplexed control



arXiv: 2312.06911

复用控制线实现并行门操作

Acknowledgement



BAQIS
wechat



Group
website



Available positions:
Research scientist / Engineer /
Postdoc / Student / Intern

Website: quantum.baqis.ac.cn
Contact: yanfei@baqis.ac.cn

Thank you!

